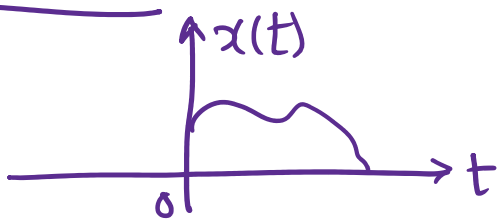
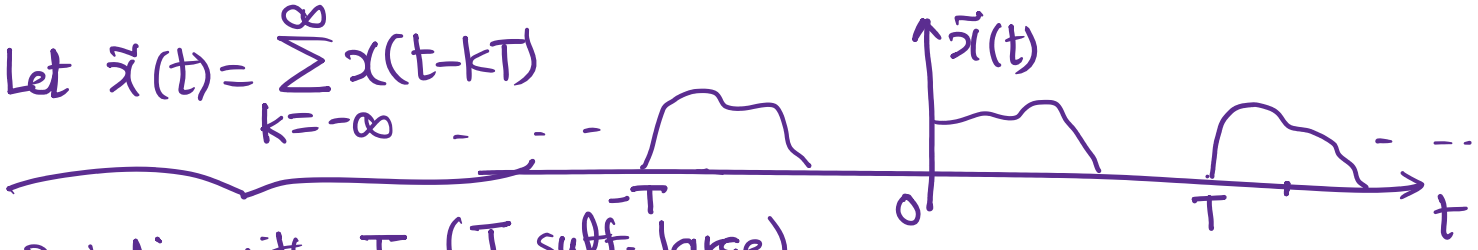
Continuous Time Fourier Transform (CTFT)

Let $x(t)$ be aperiodic signal



Let $\tilde{x}(t) = \sum_{k=-\infty}^{\infty} x(t-kT)$



Periodic with T (T suff. large)

$$\Rightarrow \quad \tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \quad , \omega_0 = \frac{2\pi}{T}$$

$$a_k = \frac{1}{T}\int_0^T \tilde{x}(t) e^{-jk\omega_0 t} dt = \frac{1}{T}\int_0^T x(t) e^{-jk\omega_0 t} dt$$

$$= \frac{1}{T}\int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t} dt$$

$$X(\omega) \triangleq \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

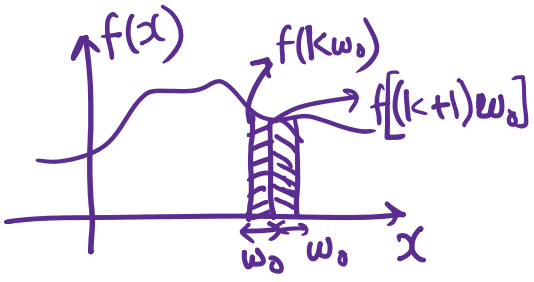
$$\Rightarrow a_k = \frac{1}{T} X(k\omega_0)$$

$$\Rightarrow \quad \tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(k\omega_0) e^{jk\omega_0 t}$$

Let $T \to \infty \Rightarrow \tilde{x}(t) = x(t)$ & $\omega_0 = \frac{2\pi}{T} \to 0$

$$\Rightarrow \quad x(t) = \frac{1}{\cancel{T}} \sum_{k=-\infty}^{\infty} X(k\omega_0) e^{jk\omega_0 t} \cdot \omega_0 \cdot \frac{\cancel{T}}{2\pi}$$

$$= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(k\omega_0) e^{jk\omega_0 t} \cdot \omega_0$$



$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$\therefore X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \longrightarrow \text{F.T.}\{x(t)\} \equiv \text{Analysis eq}^n$$

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$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t}\, d\omega \longrightarrow IFT\{x(t)\} \equiv \text{Synthesis eq}^n$$

Dirichlet Cond^n (Suff. but not Necessary)

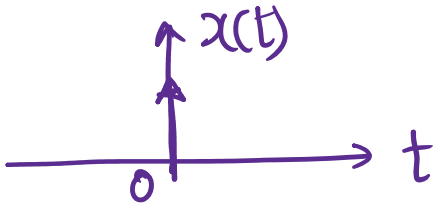
1) $\int_{-\infty}^{\infty} |x(t)|\, dt < \infty$

2) $x(t)$ has finite no. of maxima & minima over any finite period

3) " " finite discont. " "

$\Longrightarrow$ $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t}\, d\omega$ except at the pt. of discont., where value = avg. of values on either sides of discont.

<u>Ex. 1</u> $x(t) = \delta(t)$



$$\Rightarrow X(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t}\, dt = 1$$

$$\therefore \delta(t) \longleftrightarrow 1$$

<u>Ex. 2</u>: $e^{-at} u(t) = x(t)$

$$X(\omega) = \int_{0}^{\infty} e^{-at} e^{-j\omega t}\, dt = \int_{0}^{\infty} e^{-(a+j\omega)t}\, dt = \frac{e^{-(a+j\omega)\infty} - 1}{-(a+j\omega)}$$

$$= \frac{1}{a+j\omega}, \quad Re\{a\} > 0$$

$$e^{-at} u(t) \longleftrightarrow \frac{1}{a+j\omega}, \quad Re\{a\} > 0$$

<u>Ex. 3</u>: $e^{-a|t|} = x(t)$

$\int_{-\infty}^{0} e^{at} e^{-j\omega t}\, dt + \int_{0}^{\infty} e^{-at} e^{-j\omega t}\, dt$

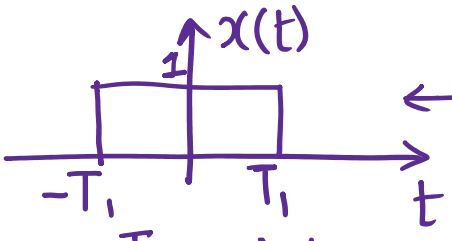
Ex. 3: $e^{-a|t|} = x(t)$

$$X(\omega) = \underbrace{\int_{-\infty}^{0} e^{at} e^{-j\omega t}\,dt}_{\frac{1}{a-j\omega},\ Re\{a\}>0} + \underbrace{\int_{0}^{\infty} e^{-at} e^{-j\omega t}\,dt}_{\frac{1}{a+j\omega},\ Re\{a\}>0}$$

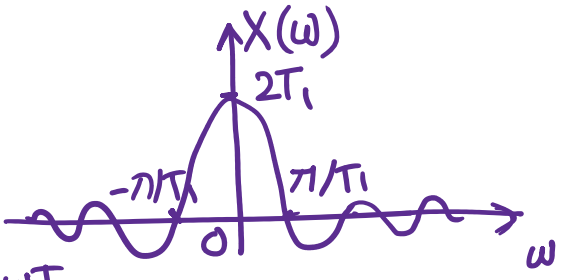
$$= \frac{1}{a-j\omega} + \frac{1}{a+j\omega},\quad Re\{a\}>0$$

$$\therefore\ e^{-a|t|} \longleftrightarrow \frac{2a}{a^2+\omega^2},\quad Re\{a\}>0$$

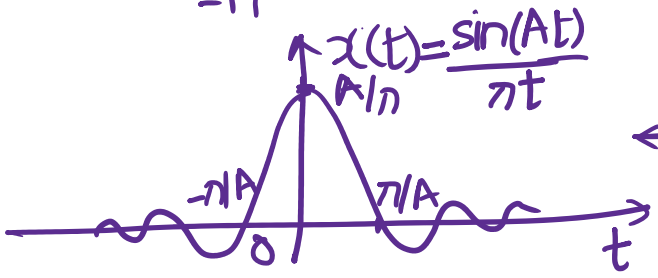
Ex. 4:



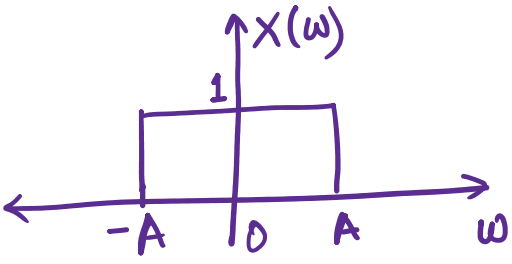
$\longleftrightarrow X(\omega)$?



$$X(\omega) = \int_{-T_1}^{T_1} 1\cdot e^{-j\omega t}\,dt = \frac{e^{-j\omega T_1} - e^{j\omega T_1}}{-j\omega} = \frac{2\sin(\omega T_1)}{\omega}$$



$\longleftrightarrow$

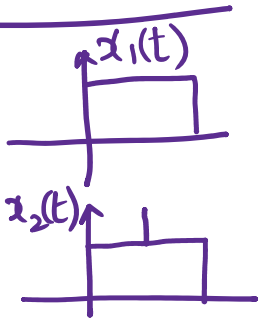


$$x(t) = \frac{1}{2\pi}\int_{-A}^{A} e^{j\omega t}\,d\omega = \frac{e^{jAt} - e^{-jAt}}{2\pi\cdot jt} = \frac{\sin(At)}{\pi t}$$

Properties:

1) Linear & Invertible

$$x(t) \longrightarrow \boxed{F.T.} \longrightarrow X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\,dt$$



$\because [x(t)] \longleftrightarrow [X(\omega)]$ are unique.

2) $x(t)$ is real $\Rightarrow X(-\omega) = X^*(\omega)$

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Pf: $X(-\omega) = \int x(t) e^{j\omega t} dt = \int x^*(t) e^{j\omega t} = X^*(\omega)$

$\therefore$ $|X(\omega)|$ is symmetric

& $\angle X(\omega)$ is antisymmetric

3) $x(t)$ is real & even $\Rightarrow$ $X(\omega)$ is real & even

Pf: $x(-t) \longleftrightarrow X(-\omega) = X^*(\omega)$

$\therefore x(t) \longleftrightarrow X(-\omega) = X(\omega) = X^*(\omega)$ $\Big\}$ $\therefore$ Proved.

4) $x(t-t_0) \longleftrightarrow X(\omega) e^{-j\omega t_0}$ $\Big\}$ <u>Time Shifting</u>

Pf: $\int_{-\infty}^{\infty} x(t-t_0) e^{-j\omega t} dt$, $t - t_0 = t'$

$$= \int_{-\infty}^{\infty} x(t') e^{-j\omega(t'+t_0)} dt' = e^{-j\omega t_0} \underbrace{\int_{-\infty}^{\infty} x(t') e^{-j\omega t'} dt'}_{X(\omega)}$$

$\therefore x(t-t_0) \longleftrightarrow X(\omega) e^{-j\omega t_0}$

5) $x(t) e^{j\omega_0 t} \longleftrightarrow X(\omega - \omega_0)$ $\Big\}$ Freq. Shifting

$$\int_{-\infty}^{\infty} x(t) e^{j\omega_0 t} e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t) e^{-j(\omega-\omega_0)t} dt = X(\omega - \omega_0)$$

6) $\frac{dx(t)}{dt} \longleftrightarrow j\omega X(\omega)$

Pf: $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$

$$\therefore \underbrace{\frac{dx(t)}{dt}}_{y(t)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \underbrace{j\omega X(\omega)}_{Y(\omega)} e^{j\omega t} d\omega$$

$\therefore \frac{dx(t)}{dt} \longleftrightarrow j\omega X(\omega)$

$$\therefore \quad \frac{dx(t)}{dt} \longleftrightarrow j\omega X(\omega)$$

7) $$tx(t) \longleftrightarrow j\frac{dX(\omega)}{d\omega}$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\Rightarrow \frac{dX(\omega)}{d\omega} = \int x(t)(-jt)e^{-j\omega t} dt = -j\int_{-\infty}^{\infty} tx(t) e^{-j\omega t} dt$$

$$\therefore j\frac{dX(\omega)}{d\omega} = \int_{-\infty}^{\infty} tx(t) e^{-j\omega t} dt$$

$$\therefore \quad tx(t) \longleftrightarrow j\frac{dX(\omega)}{d\omega}$$

8) $$x(at) \longleftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

$$\int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt = \underbrace{\int_{-\infty}^{\infty} x(t') e^{-j\omega \frac{t'}{a}} \frac{dt'}{a}}_{a>0}, \quad t' = at$$

$$= \frac{1}{a}\int x(t') e^{-j\frac{\omega}{a}t'} dt' = \frac{1}{a} X\left(\frac{\omega}{a}\right) \quad , a>0$$

$$a<0 \Rightarrow \frac{-1}{a} X\left(\frac{\omega}{a}\right)$$

$$\therefore x(at) \longleftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$